Page No.:
Date: 26-2-26

Assignment - 3

Q1. Double Number Pyramid

For n = 4

```
1
1 2
1 2 3
1 2 3 4
1 2 3
1 2
1
```

* Dry Run

n = 4

i = 1 ; i <= n
↳ j = 1 ; j <= i
↳ 1

i = 2
↳ j <= i
↳ 1 2

i = 3
↳ j <= i
↳ 3

i = n-1 ; i >= 1
↳ j = 1 ; j <= i
↳ 1 2 3

```
1
1 1
1 1 2
1 2 3
1 2 3 4
```

Page No.:
Date:

Number Butterfly

Q2

```
1       1
1 2     2 1
1 2 3   3 2 1
1 2 3 4 4 3 2 1
1 2 3 4 4 3 2 1
1 2 3   3 2 1
1 2     2 1
1       1
```

* Dry Run n = 4

* 1, 0L

i = 1 ; i <= n ;
↳ j = 1 ; j <= i
↳ 1

2 <= n

j <= i
↳ 1 2

* ~~right~~ spacing

for 1 ; j <= 2*(n-1)
↳ . ..

Right
* j = i ; j >= 1
↳ 1

i >= 1

2 1

Q3 Cross (x) Number Pattern

```
1   1
  2 2
    3
  4 4
  5 5
```

n = 5

i = 1; i<=n
↳
j = 1; j<=n
↳ if(j==i || j==n-i+1){
↳ " "

j<=n
↳ (else) " " 2 " " 2

Q4 Spiral Loop Pattern

```
3 3 3 3 3
3 2 2 2 3
3 2 1 2 3
3 2   2 3
3 3 3 3 3
```

• Dry Run n = 3;

i = 1 ; i<= 2*n-1
↳ j = 1; j<= 2*n-1
↳

Q5 Pascal triangle

Dry run n = 5

```
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
```

i = 0; i<n; i
↳ num 1
↳ j = 0; j<=i; j
↳ n+" "
n = num*(i-j)/(j+1);
↳ 1
j<=i
↳ 1 1

Q6 Diamond with continous numbers

n = 4

```
   1
  2 3
 4 5 6
7 8 9 10
 11 12 13
  14 15
   16
```

m = 4 , num = 1

Dry run

i = 1 ; i <= n
↳ j = 1 ; j <= i
↳ num ↳ 1
num++

2 <= i
↳ num ↳ 2 3
num++

i = n-1 ; i >= 1
↳ j = 1 ; j <= i
↳ num
num++

Q7. Binary Diamond

```
   1
  0 1
 1 0 1
0 1 0 1
 1 0 1
  0 1
   1
```

Dry Run n = 4

i = 1 ; i <= n
↳ val = i % 2
↳ j = 1 ; j <= i
↳ val = 1 - val
1 + 1 = 1 - 1

2 <= n
↳ i <= i
↳ val = 2 % 2 = 0
↳ 0
= 1 % 2 = 1